

Multi-config experiment report

27 January 2026

Overview

This report summarizes the controlled experiments with various configuration for kernels. All runs are logistic models with different kernel counts and selection method.

Run configuration

Common settings: topM=300, response_fn=abs_max, batch size 64, lr=5e-4, epochs=60.

MMR runs

Run	Model	K	topM	std_feat	std_resp
exp_022_mmr	logistic	3	300	False	False
exp_023_mmr	logistic	5	300	False	False
exp_024_mmr	logistic	200	300	False	False
exp_025_mmr	logistic	300	300	False	False

ABESS runs

Run	Model	K	topM	std_feat	std_resp
exp_029_abess	logistic	3	300	False	False
exp_030_abess	logistic	5	300	False	False
exp_031_abess	logistic	200	300	False	False
exp_032_abess	logistic	300	300	False	False

Validation/Test metrics

MMR runs

Run	Val AUC	Val ACC	Test AUC	Test ACC	Intercept
exp_022_mmr	0.9085	0.8011	0.9239	0.7914	-0.691
exp_023_mmr	0.9071	0.7765	0.9234	0.7594	-0.908
exp_024_mmr	0.9047	0.8156	0.9180	0.7909	-1.215
exp_025_mmr	0.8790	0.8006	0.8927	0.8182	-1.204

ABESS runs

Run	Val AUC	Val ACC	Test AUC	Test ACC	Intercept
exp_029_abess	0.9059	0.7615	0.9216	0.7455	-0.649
exp_030_abess	0.9075	0.7838	0.9225	0.7684	-1.004
exp_031_abess	0.9095	0.7849	0.9236	0.7690	-1.267
exp_032_abess	0.8824	0.8028	0.8966	0.8032	-1.224

Heatmap comparisons

MMR

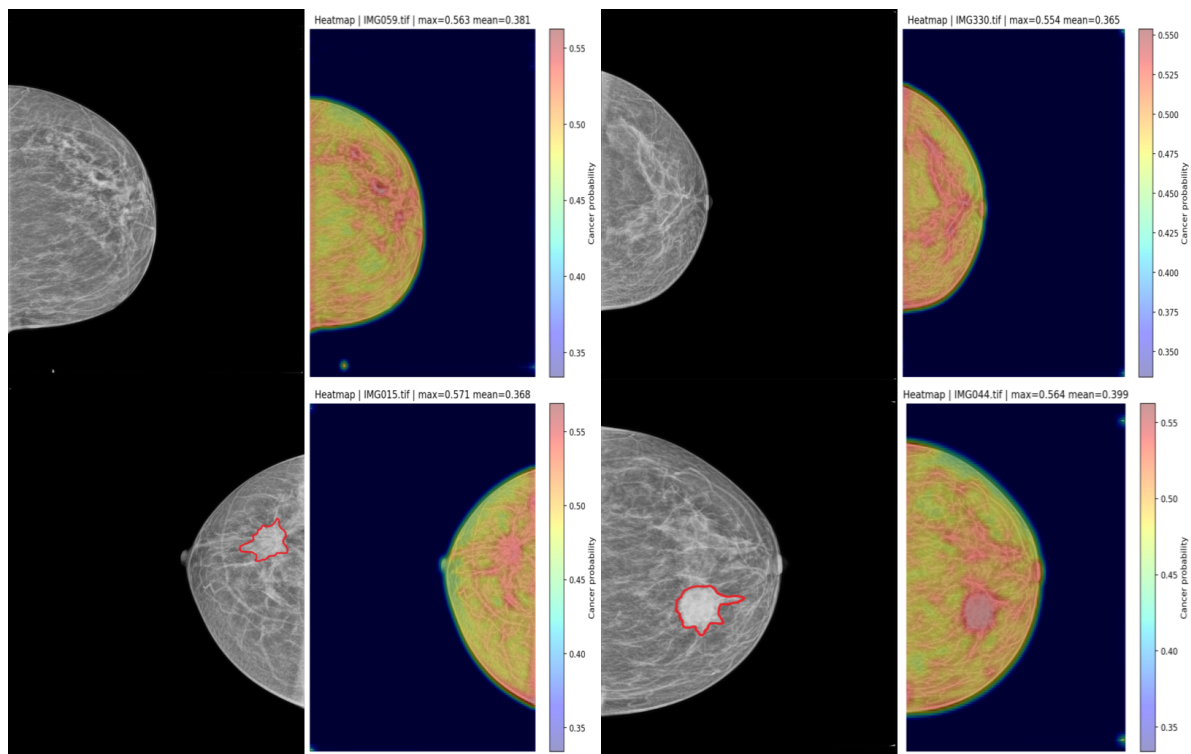


Figure 1: MMR heatmaps for exp_022 (2 normal + 2 malignant).

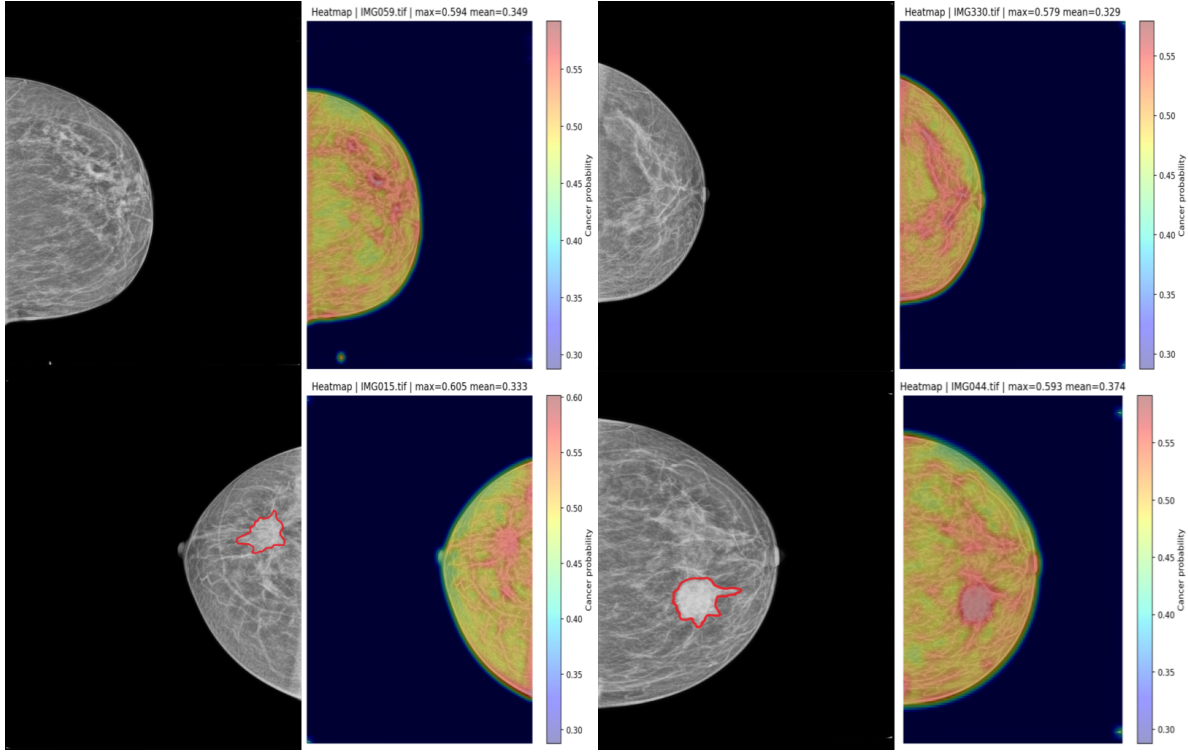


Figure 2: MMR heatmaps for exp_023 (2 normal + 2 malignant).

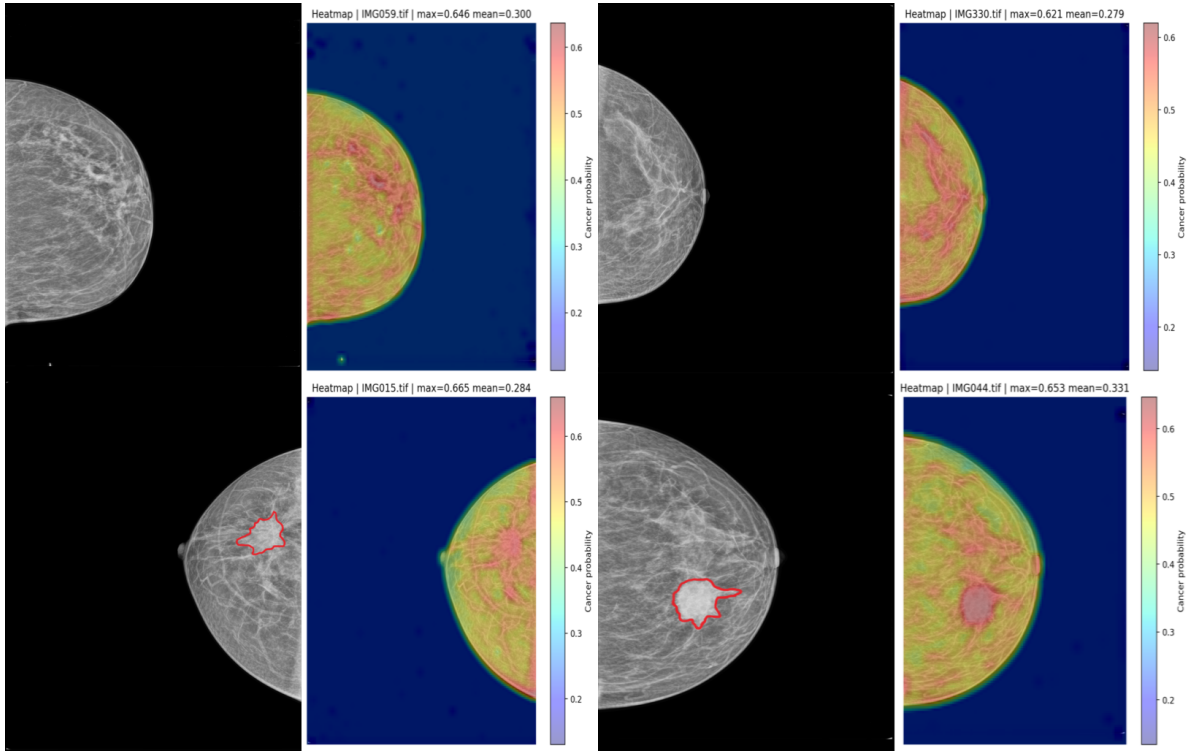


Figure 3: MMR heatmaps for exp_024 (2 normal + 2 malignant).

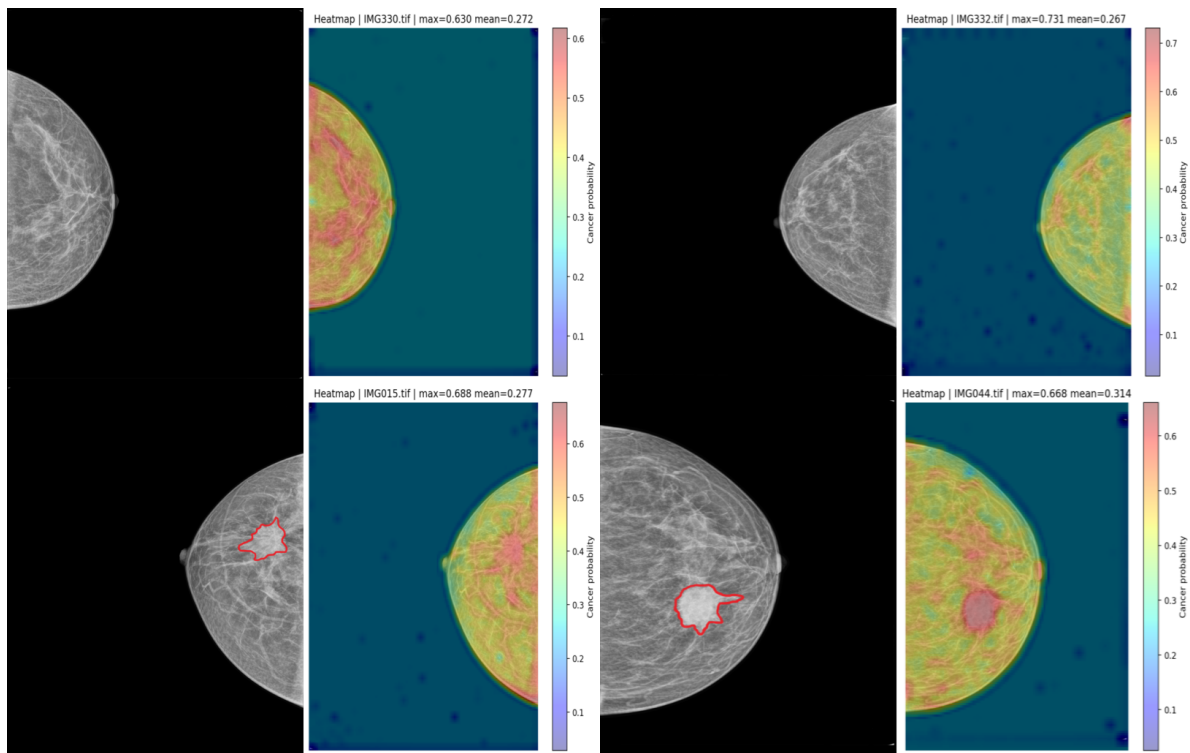


Figure 4: MMR heatmaps for exp_025 (2 normal + 2 malignant).

ABESS

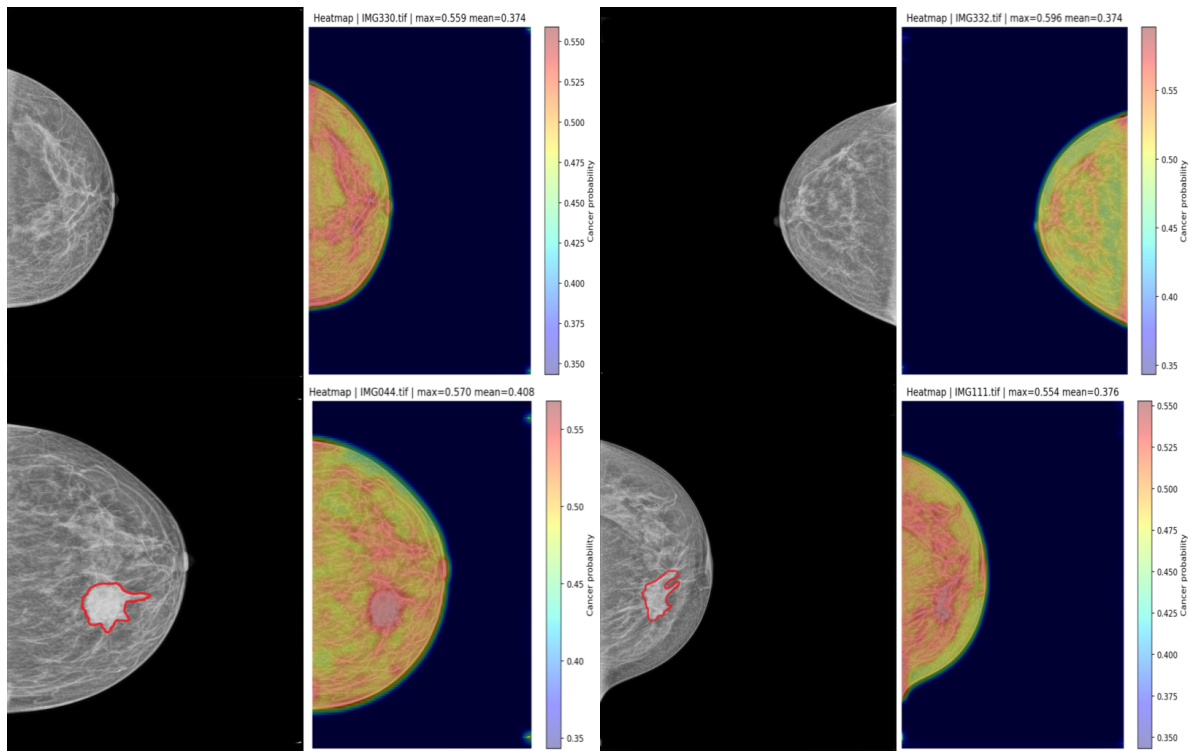


Figure 5: ABESS heatmaps for exp_029 (2 normal + 2 malignant).

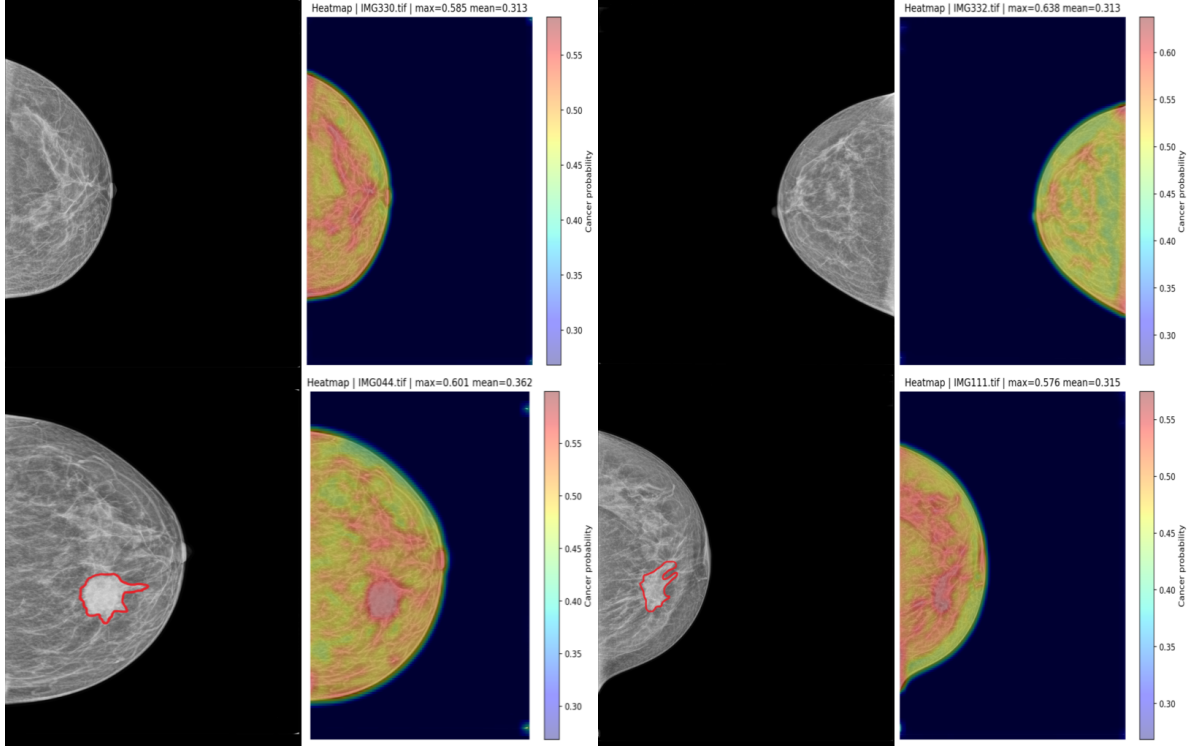


Figure 6: ABESS heatmaps for exp_030 (2 normal + 2 malignant).

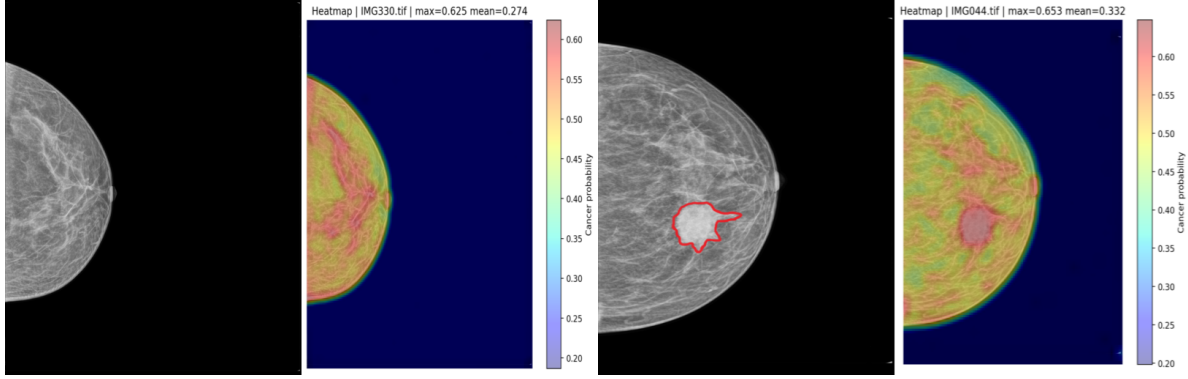


Figure 7: ABESS heatmaps for exp_031 (partial set available).

Summary

Across MMR runs, test AUC is highest for exp_022_mmr (0.9239), while test accuracy peaks at exp_025_mmr (0.8182). Across ABESS runs, exp_031_abess achieves the best test AUC (0.9236), and exp_032_abess has the best test accuracy (0.8032). Overall, MMR runs show slightly higher peak AUC/ACC than ABESS in these settings.

Conclusion

If prioritizing test AUC, exp_022_mmr is the best overall. If prioritizing test accuracy, exp_025_mmr is the best overall. Among ABESS configurations, exp_031_abess (AUC) and exp_032_abess (ACC) are the

strongest.

Summary plot

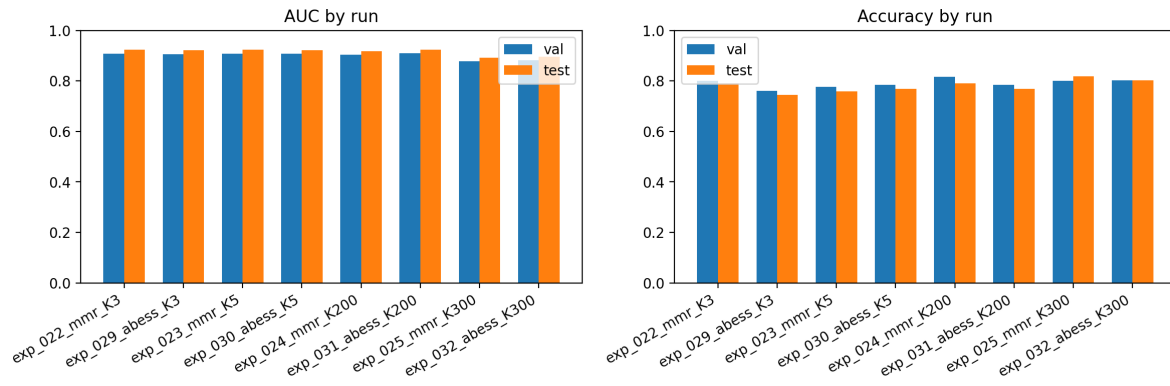


Figure 8: AUC and accuracy for validation/test across multi-config runs.

Additional MMR variants (n_per_family / K)

Run	n_per_family	K	topM	Val AUC	Test AUC	Test ACC
exp_022_mmr_n1_k1	1	1	300	0.8994	0.9157	0.8449
exp_022_mmr_n3_k1	3	1	300	0.9010	0.9179	0.8428
exp_022_mmr_n3_k3	3	3	300	0.9085	0.9101	0.8417
exp_022_mmr_n10_k3	10	3	65	0.9054	0.9032	0.8449

Summary These reduced-bank variants maintain strong test accuracy (around 0.84–0.85) but slightly trail the best baseline MMR AUCs. Compared with the top-of-report baselines (exp_022_mmr AUC 0.9239, exp_025_mmr ACC 0.8182), the n_per_family variants trade a small AUC drop for competitive accuracy.

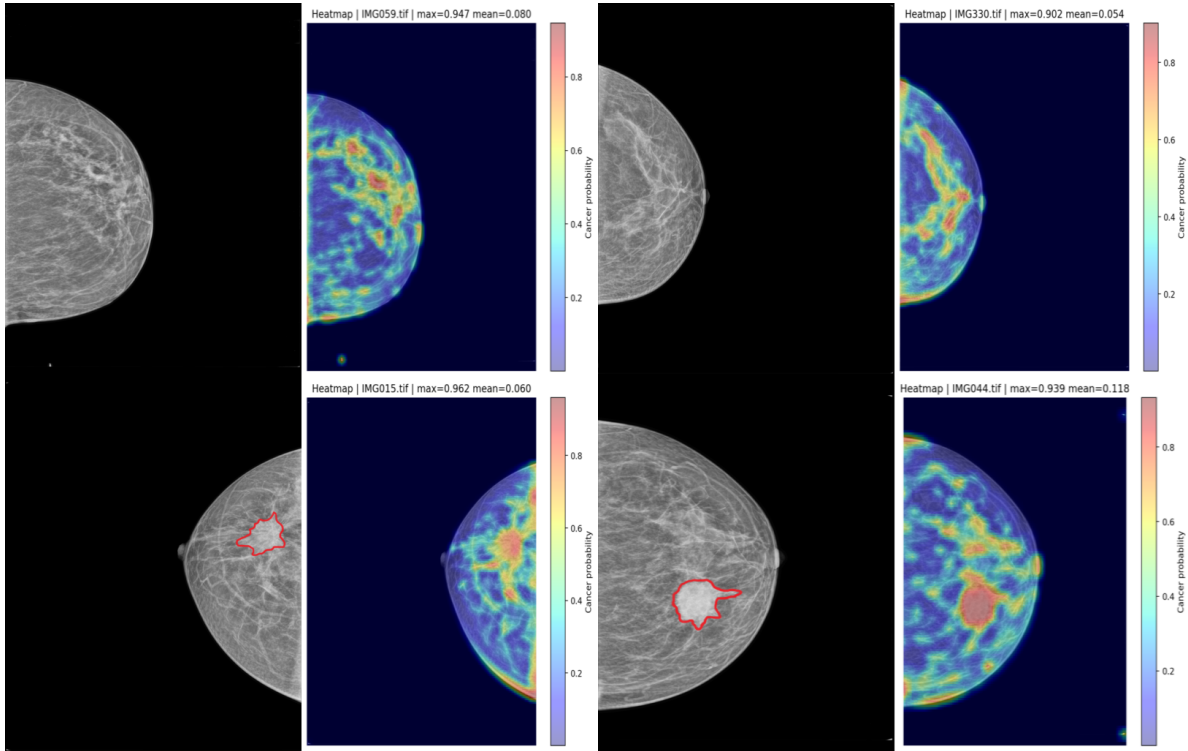


Figure 9: Heatmaps for exp_022_mmr_n1_k1.

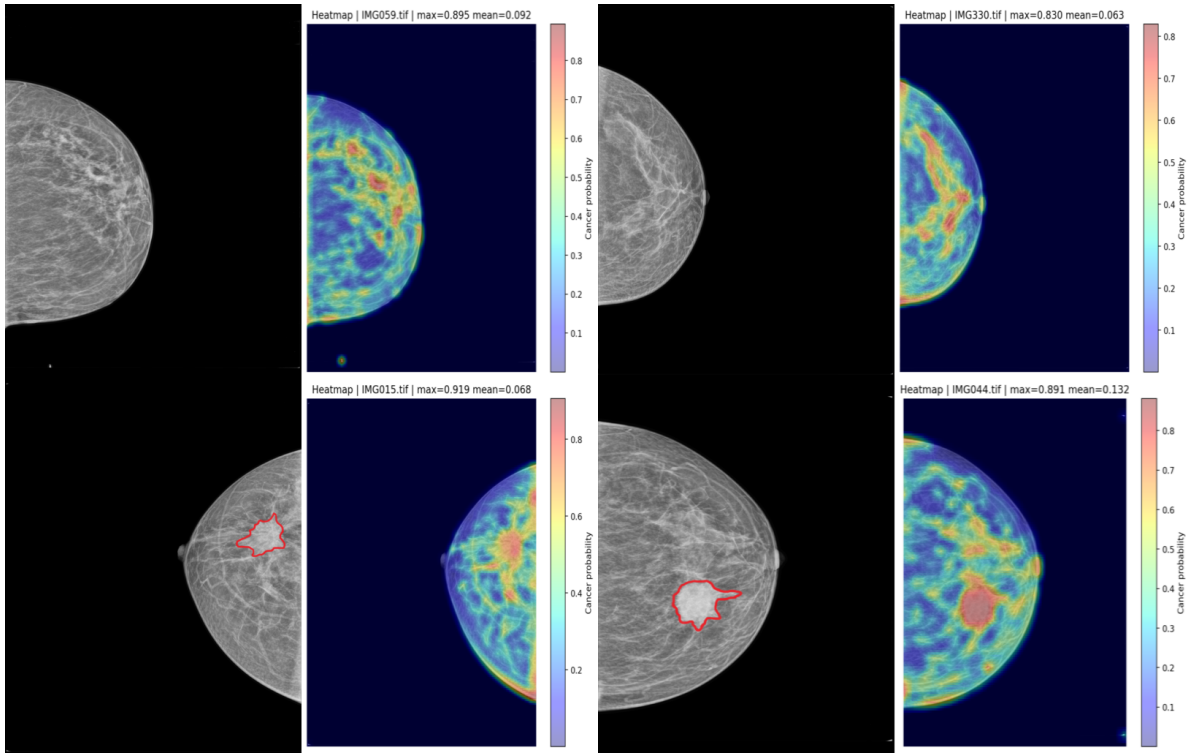


Figure 10: Heatmaps for `exp_022_mmr_n3_k1`.

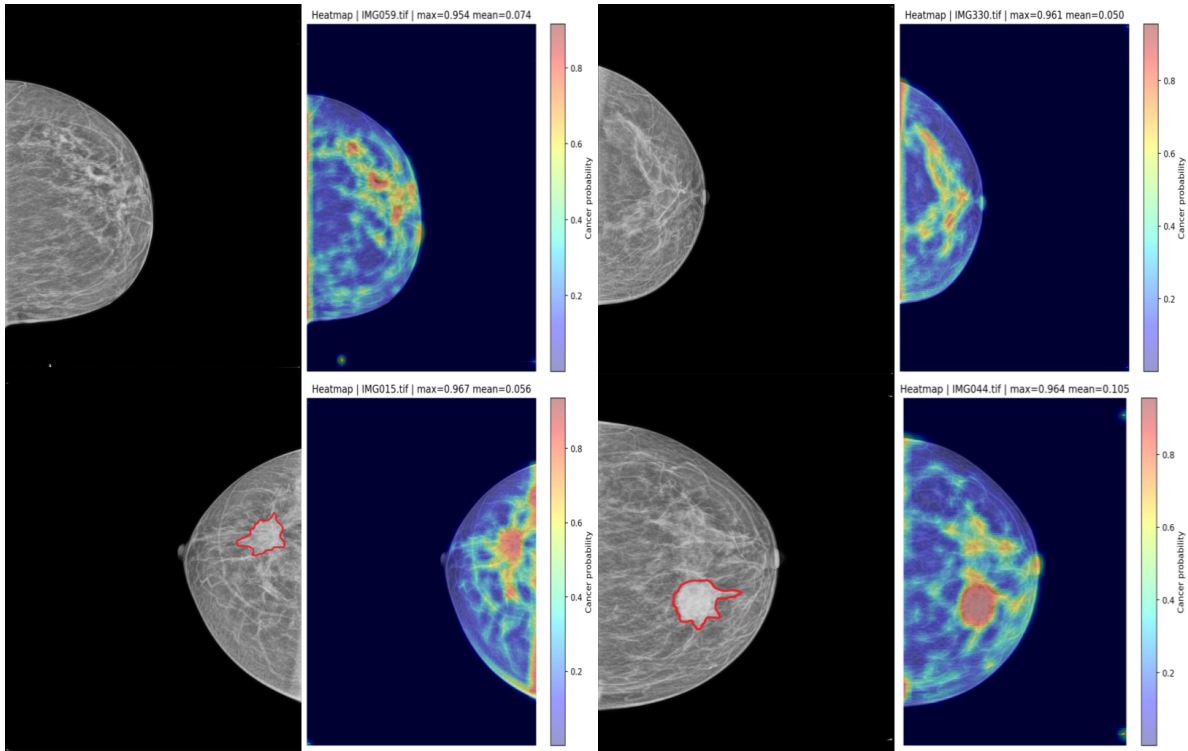


Figure 11: Heatmaps for `exp_022_mmr_n3_k3`.

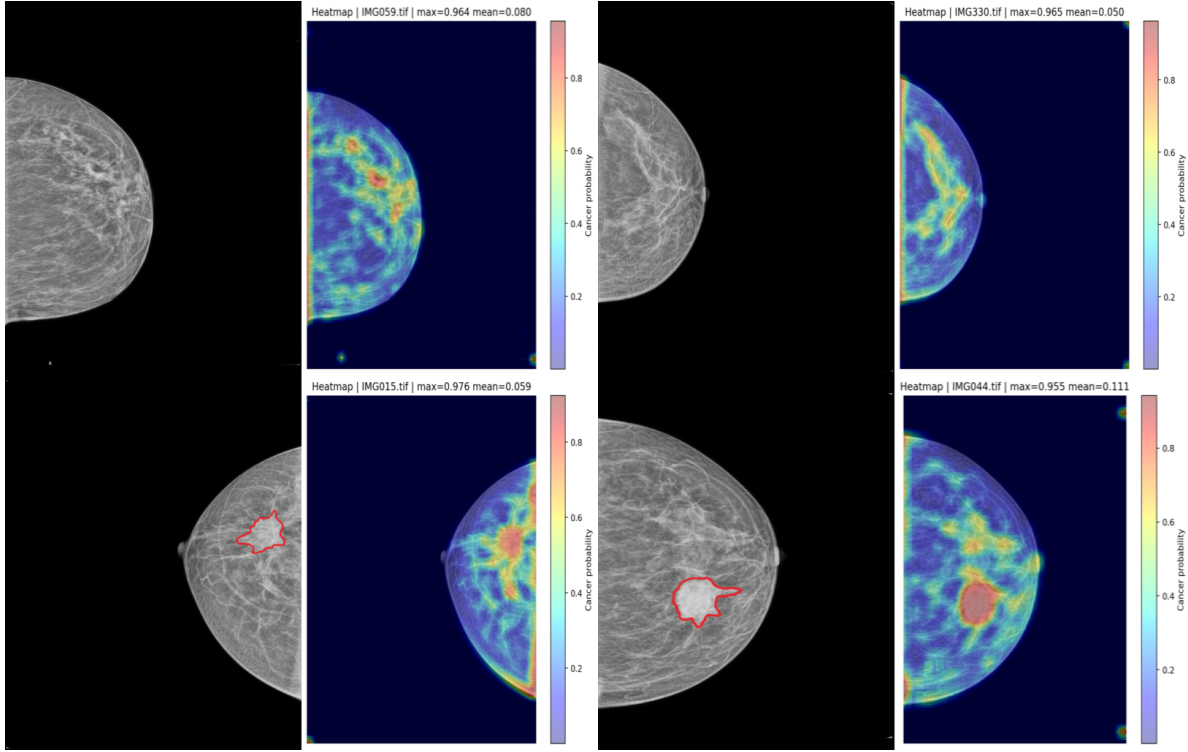


Figure 12: Heatmaps for `exp_022_mmr_n10_k3`.

Composition experiments

In this section, we explore two new runs with composite kernels using different kernel bank sizes. We vary the number of kernels per family (`n_per_family`) to create banks of different sizes, and select a number of kernels (`n_Composition`) for composition of the final model. In both cases, we use AUC to weight the composite kernels. Also experimented with fisher weighting, but results were slightly worse.

Shared settings:

- Selection method: MMR
- Model: Logistic
- `weight_method`: AUC
- `normalize`: None

Run	<code>n_per_family</code>	<code>n_Composition</code>	Val AUC	Test AUC	Test ACC
<code>exp_022_mmr_nperFamily10_nComposition25</code>	10	65	0.9049	0.9096	0.8487
<code>exp_022_mmr_nperFamily50_nComposition125</code>	50	325	0.9217	0.9260	0.8652

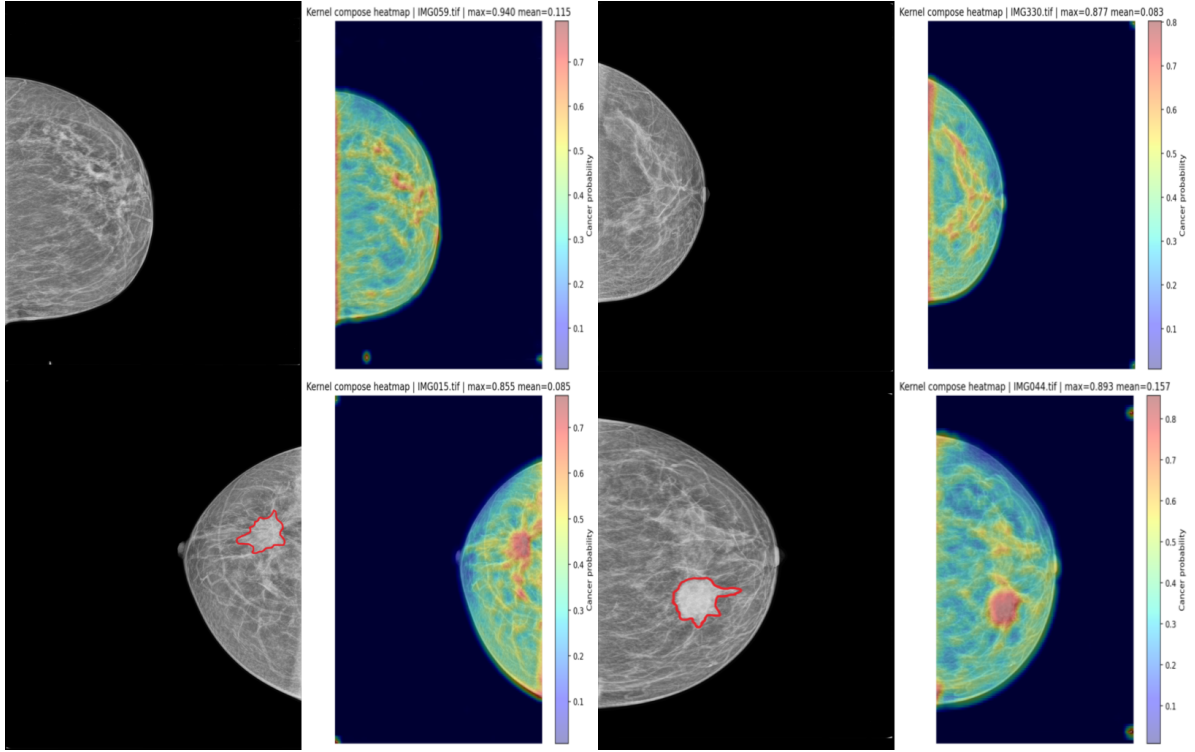


Figure 13: Composition heatmaps for exp_022_mmr_nperFamily10_nComposition25.

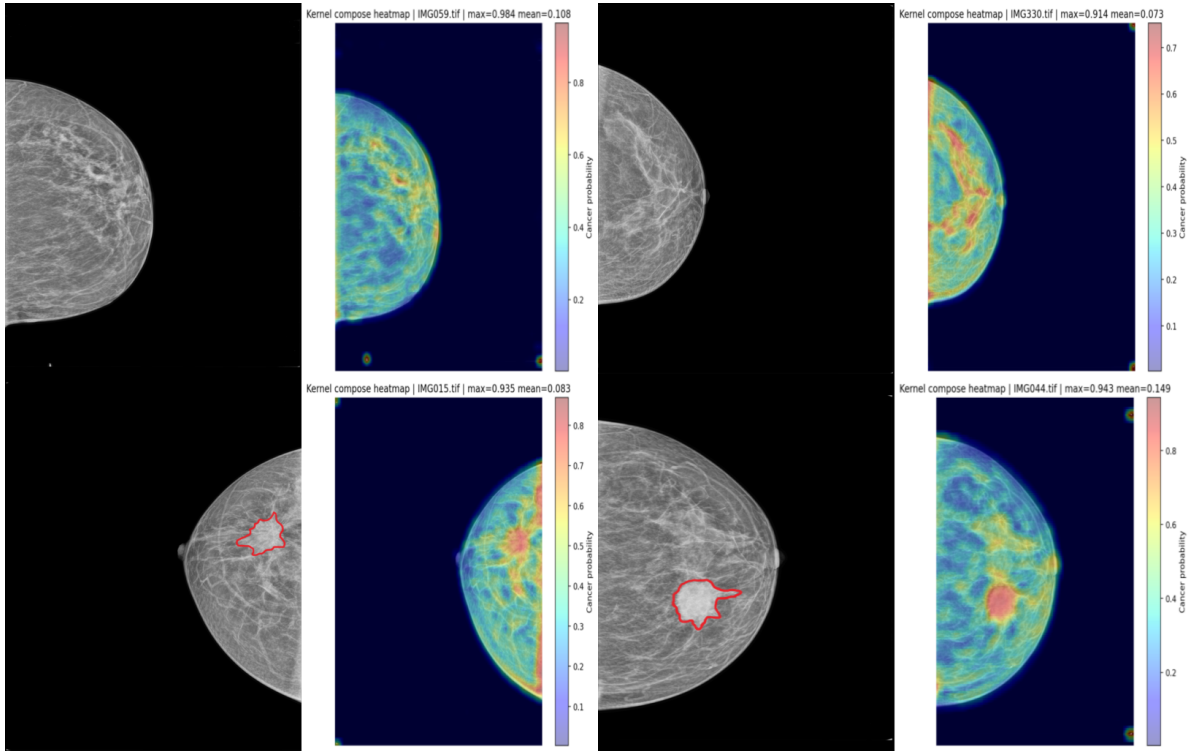


Figure 14: Composition heatmaps for exp_022_mmr_nperFamily50_nComposition125.

Conclusion The 125-composition run is the strongest of the new additions, exceeding the earlier baselines in both test AUC (0.9260 vs 0.9239 for exp_022_mmr) and test accuracy (0.8652 vs 0.8182 for exp_025_mmr). The 25-composition run is competitive but does not surpass the best baseline.